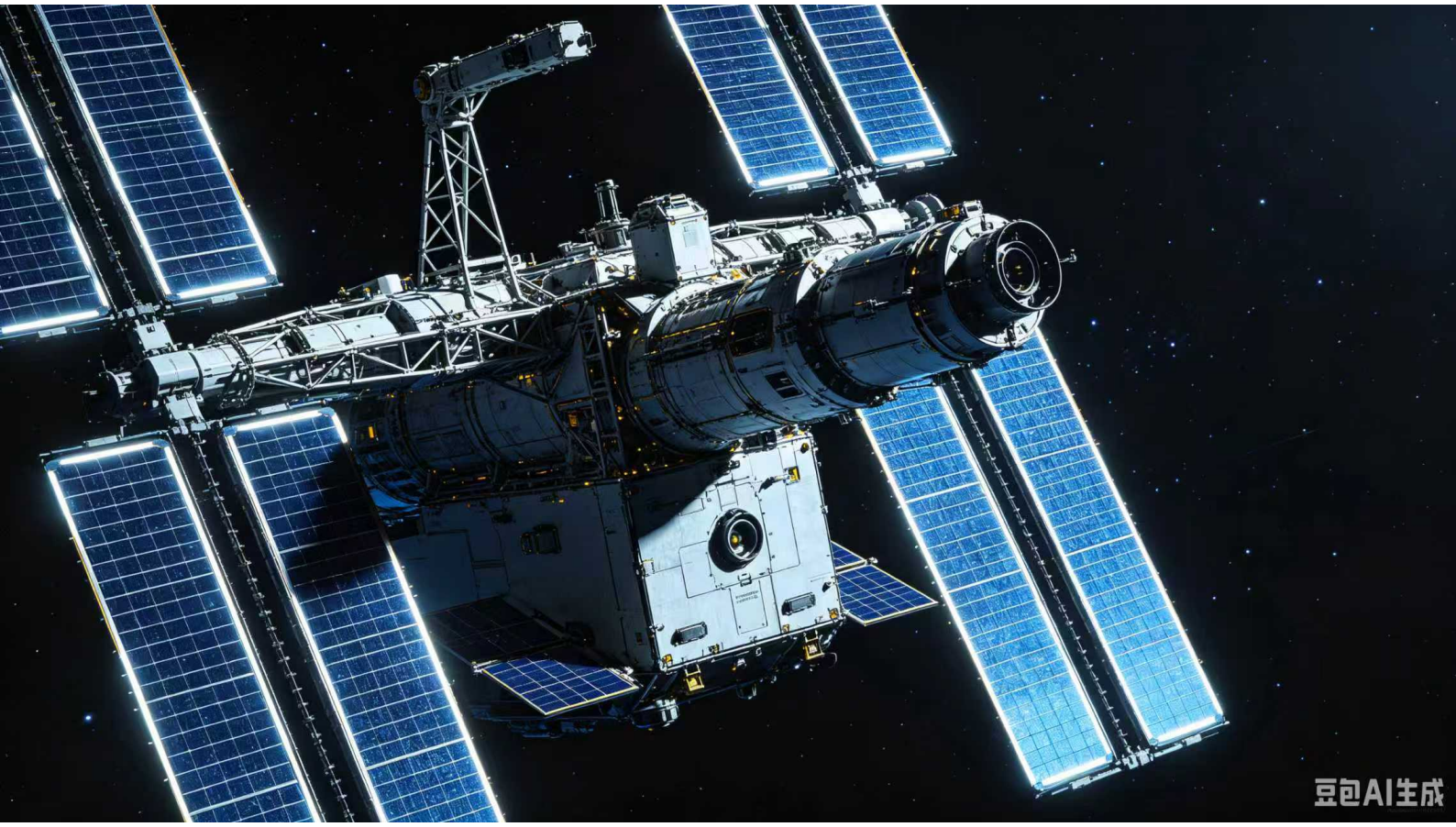


11. Faraday's Law of Induction

Our virtues and our failings are inseparable, like force and matter.
When they separate, man is no more.

– *Nikola Tesla, "The Problem of Increasing Human Energy," Century Illustrated Magazine*, June 1900

From statics to dynamics. In electrostatics and magnetostatics, the fields obey:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \qquad \nabla \times \mathbf{E} = 0,$$
$$\nabla \cdot \mathbf{B} = 0, \qquad \nabla \times \mathbf{B} = \mu_0 \mathbf{j}.$$

The discovery of electromagnetic induction revealed that a changing magnetic field produces an electric field, modifying $\nabla \times \mathbf{E} = 0$ into a dynamic equation. This chapter establishes the link from a changing $\mathbf{B}$ to an $\mathbf{E}$; the other direction, from a changing $\mathbf{E}$ to a $\mathbf{B}$, is Maxwell's displacement current (Chapter 14chapter.14), and the two together led Maxwell to the unification of electricity and magnetism.

11.1 The Discovery of Induction

In 1831 Michael Faraday discovered electromagnetic induction; his first paper on it was read to the Royal Society that November and printed in 1832. Two simple experiments show what he found. Suppose we set up a conducting loop L connected to a sensitive ammeter and do these experiments (Figs. 11.1figure.11.1 and 11.2figure.11.2):

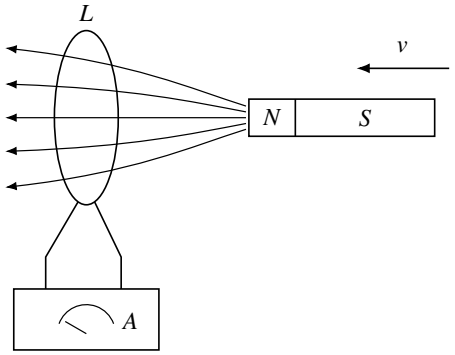


Figure 11.1: A magnet moving near a loop.

Experiment 1: We use a bar magnet to do the experiment.

1. If we move a bar magnet toward the loop, a current suddenly appears in the circuit.

2. The current disappears when the magnet stops.

3. If we then move the magnet away, a current again suddenly appears, but now in the opposite direction.

4. Faster motion produces greater current.

5. The north pole and the south pole produce opposite currents when the same experiment is conducted.

Experiment 2: We now put a second conducting loop L' connected to a battery beside the first loop L.

1. If we turn on a current in L', the ammeter suddenly and briefly registers an induced current in L.

2. If we turn off the current in L', another induced current in the opposite direction is briefly and suddenly registered.

3. The current in L is only obtained when the current in L' is changing. There is no current in L if the current in L' is constant (even if it is large).

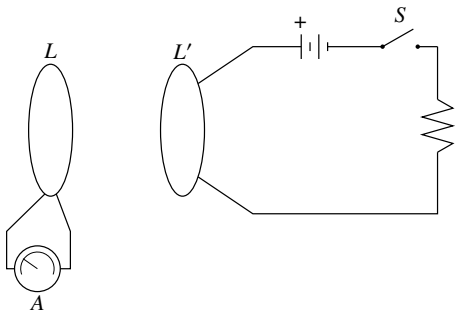


Figure 11.2: A current switched on and off near a loop.

Faraday realized via these two experiments that an emf and a current can be induced in a loop, only by changing the amount of the magnetic field passing through the loop — the **magnetic flux**. Faraday's law of induction therefore states that an emf is induced in a closed loop if the magnetic flux through the loop is changing. Mathematically, it is expressed as:

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}\int \mathbf{B}\cdot d\mathbf{S}$$

The unit of magnetic flux is the weber, 1 Wb = 1 T m^2. The sense of $\mathcal{E}$ counted as positive and the normal $d\mathbf{S}$ are related by the right-hand rule: with the fingers of the right hand curled along the positive sense round the loop, the thumb gives the direction of $d\mathbf{S}$. For a coil of N turns, each linking the same flux Φ_B, the total induced emf is

$$\mathcal{E} = -N\frac{d\Phi_B}{dt}$$

11.2 Lenz's Law

Experimentally, an induced current has a direction such that the magnetic field due to the current *opposes* the change in the magnetic flux that induces the current. More specifically, if we look at the first set of experiments done by Faraday, it says:

1. If the north pole of the magnet approaches the loop from the right, the magnetic flux through the loop (pointing to the left, away from the magnet) increases. Thus, the magnetic field due to the induced current must point to the right to oppose such an increase. Therefore, the current in the loop must be clockwise as seen by an observer on the left of the loop.

2. If the north pole of the magnet is pulled away from the loop toward the right, the magnetic flux through the loop (pointing to the left, away from the magnet) decreases. Thus, the magnetic field due to the induced current must point to the left to oppose such a decrease. Therefore the current in the loop must be counter-clockwise as seen by an observer on the left of the loop.

The other cases follow the same two steps; work one out to check that you have the rule.

Lenz's law says that the induced current opposes the change in the magnetic flux; when that change is caused by motion, the current therefore resists the motion. The law is a consequence of energy conservation.

Example 11.1: A circular loop in a varying magnetic field.

We assume the magnetic field $\mathbf{B} = \hat{z}B_0 \cos \omega t$ fills a long cylindrical region, coaxial with the loop. The circular loop of radius R lies in the xy-plane (Fig. 11.3figure.11.3), with $d\mathbf{S}$ along $+\hat{z}$. Thus, the magnetic flux through the loop is:

$$\int \mathbf{B} \cdot d\mathbf{S} = B_0 \pi R^2 \cos \omega t$$

Thus we have $\mathcal{E} = \omega B_0 \pi R^2 \sin \omega t$ for the circular loop.

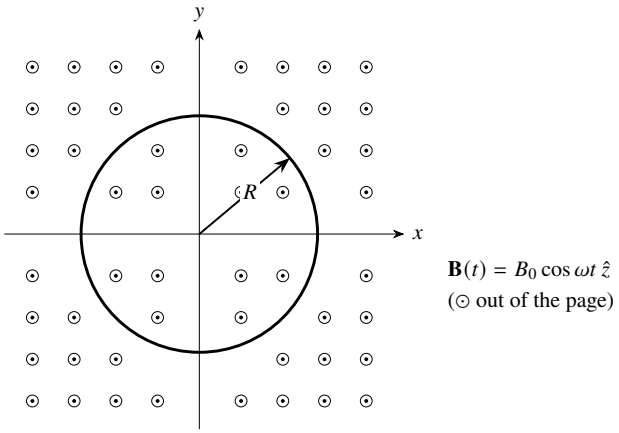


Figure 11.3: A circular loop of radius R in a time-varying magnetic field $\mathbf{B}(t) = B_0 \cos \omega t\, \hat{z}$ pointing out of the page. The induced emf is $\mathcal{E} = -d\Phi_B/dt = \omega B_0 \pi R^2 \sin \omega t$.

Example 11.2: A sliding conducting rod on two connected parallel rails.

Suppose the parallel rails, as shown in Fig. 11.4figure.11.4, are separated by a distance L in the xy-plane, and there is a uniform magnetic field $\mathbf{B} = B_0\hat{z}$. The velocity of the sliding rod is v. Assuming the distance between the rod and the connected ends of the rails is x, the flux is:

$$\int \mathbf{B} \cdot d\mathbf{S} = B_0 L x$$

Therefore, the emf induced in the loop is:

$$\mathcal{E} = -B_0 L \frac{dx}{dt} = -B_0 L v$$

Its magnitude is $|\mathcal{E}| = B_0 L v$; the minus sign fixes the sense of the induced current, as required by Lenz's law.

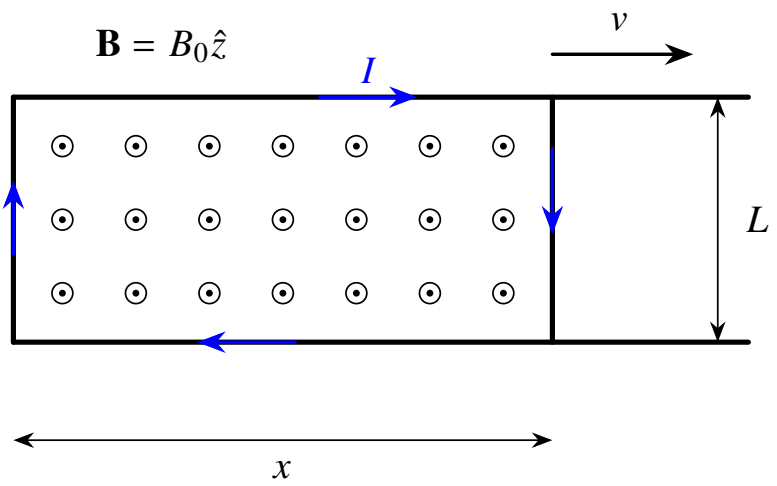


Figure 11.4: A conducting rod slides with velocity v on two parallel rails connected at one end, in a uniform magnetic field $\mathbf{B} = B_0\hat{z}$ pointing out of the page. The changing flux induces an emf $\mathcal{E} = -B_0 L v$; the blue arrows give the direction of the induced current.

11.3 Induced Electric Fields

We see that an emf will be induced if the magnetic flux is varying inside a loop. The question now is whether there is an electric field associated with this emf.

This is indeed a tricky question. Let us distinguish 3 situations.

1. We look at Example 11.2example.11.2, the sliding conducting rod on two parallel rails. If we think carefully about this case, the driving force of the emf in this case is indeed the Lorentz force on the charge carriers inside the rod. When the rod slides along the rails, it takes the charge carriers along with it, which then feel a Lorentz force of magnitude $F = qvB$. However, the carriers are constrained by the geometry and can only move along the rod. Therefore, with the sense of circulation fixed by $d\mathbf{S} = +\hat{z}$, the emf is $\mathcal{E} = -FL/q = -vBL$, exactly the value we are expecting. It is therefore evident that there is no *induced* electric field in this case: the field is static, so no circulating electric field is produced; the carriers are pushed along the rod by the magnetic force.

2. A magnet moving toward a loop at rest. The loop's carriers are at rest, so no magnetic force acts on them; yet a current flows, so there must be an electric field. Here the magnetic field at each point of the loop does change as the magnet approaches.

3. Finally, we take Example 11.1example.11.1. In this case, the loop and its carriers are at rest initially, therefore there must also be an electric field associated with the emf.

We see that the difference between the first case and the other two is that the magnetic field is static in the first case. We therefore arrive at an inspiring statement:

A changing magnetic field produces an electric field.

Therefore, for a loop C at rest we rewrite Faraday's induction law as

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = \mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}\int \mathbf{B} \cdot d\mathbf{S}$$

For a moving loop, such as the rod of Example 11.2example.11.2, the emf is instead the circulation of $\mathbf{E} + \mathbf{v} \times \mathbf{B}$, which equals $-d\Phi_B/dt$. When $\mathbf{B}$ is static, as here, $\mathbf{E}$ contributes nothing to it,

and all of $-d\Phi_B/dt$ comes from $\mathbf{v}\times\mathbf{B}$. For a static field the closed integral vanishes, and that is what allows the potential difference $V_b - V_a = -\int_a^b \mathbf{E}\cdot d\mathbf{l}$ to be defined; when $\mathbf{B}$ varies in time the integral instead equals the induced emf. Again, the integration on the left takes place along a closed loop C, whereas the integration on the right takes place on any surface bounded by C. Figure 11.5figure.11.5 shows the induced electric field along a circular path inside a region of varying magnetic field; no conductor is present.

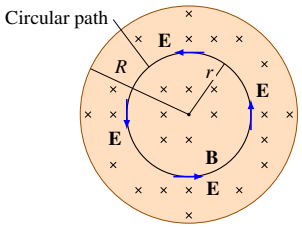


Figure 11.5: Induced electric field on a circular path of radius r inside a region where the magnetic field, directed into the page, is increasing. No conductor is present; while dB/dt is constant, the field does work $q\mathcal{E}$ on a charge carried once around the path.

11.4 Faraday's Law in the Differential Form

For a loop C and a surface S at rest, Stokes' theorem, $\oint_C \mathbf{E}\cdot d\mathbf{l} = \int_S (\nabla\times\mathbf{E})\cdot d\mathbf{S}$, turns Faraday's law into $\int_S (\nabla\times\mathbf{E} + \partial\mathbf{B}/\partial t)\cdot d\mathbf{S} = 0$. Since this holds for every such surface, the integrand vanishes:

$$\nabla\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t}$$

Now we see that the scalar potential defined previously can no longer describe the whole field once $\mathbf{B}$ varies in time, because $\nabla\times\mathbf{E}$ no longer vanishes.

Loss of electric potential. In electrostatics, $\nabla\times\mathbf{E} = 0$ guarantees that $\mathbf{E}$ is conservative and the line integral $\int \mathbf{E}\cdot d\mathbf{l}$ is path-independent, allowing the definition of a scalar potential V via $\mathbf{E} = -\nabla V$. In electrodynamics, however,

$$\oint \mathbf{E}\cdot d\mathbf{l} = -\frac{d\Phi_B}{dt} \neq 0 \qquad \text{(for a loop that encloses a changing flux)},$$

so $\mathbf{E}$ is no longer conservative: the line integral of $\mathbf{E}$ between two points depends on the path. No potential V with $\mathbf{E} = -\nabla V$ exists in a region where $\mathbf{B}$ is changing, where $\nabla\times\mathbf{E} \neq 0$, and none can be single-valued round a closed path that encircles changing flux. Motional emf is the exception: there the flux changes while $\mathbf{B}$ is static, and $\mathbf{E}$ remains conservative. The magnetic force still does no work on the carriers; it redirects them, and the energy comes from whatever moves the conductor.

Relativistic origin of magnetic fields. In special relativity, electric and magnetic fields are different manifestations of the same electromagnetic field. If two oppositely charged sheets produce an electric field E in frame F, then in a frame F' moving perpendicular to $\mathbf{E}$ at speed v, the moving charges constitute current sheets that produce a magnetic field $\mathbf{B}' = -\gamma\,\mathbf{v}\times\mathbf{E}/c^2$, of magnitude $B' = \gamma v E/c^2$, while the electric field is strengthened to $E' = \gamma E$ (here $\mu_0\epsilon_0 = 1/c^2$, with c the speed of light, and $\gamma = 1/\sqrt{1 - v^2/c^2}$). Thus a purely electric field in one frame appears as a combination of $\mathbf{E}'$ and $\mathbf{B}'$ in another. The six components of $\mathbf{E}$ and $\mathbf{B}$ together form an antisymmetric second-rank tensor under Lorentz transformations.

11.5 Exercises

1. An infinitely long conducting wire carries a current I, and a rectangular loop lies in the same plane as the wire and moves away from it perpendicular to the wire at constant speed v, as shown in Fig. 11.6figure.11.6. The loop has side a parallel to the wire, width b, and its near side is a distance c from the wire. Calculate the induced emf in the rectangle.

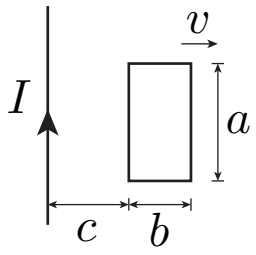


Figure 11.6: A rectangular loop leaving an infinitely long conducting wire.

2. An equilateral triangular loop of side a and resistance R is rotating about its edge 1–2 with constant angular velocity ω. A uniform magnetic field B points into the page. At time $t = 0$ the triangular plane is perpendicular to the magnetic field, as shown in Fig. 11.7figure.11.7. Take the resistance of the loop to be uniform, $R/3$ per side. The magnetic field is static, so the electric potential is well defined even though the flux through the loop is changing; across a side that carries a motional emf e and a current I, the potential difference between its ends is $V_a - V_b = e - IR/3$, where e and I are both measured positive in the same sense along the side (on the side with $e = 0$, take that sense to be the direction of I) and a is the end toward which that sense points. Calculate the electric potential difference between each pair of vertices at any time t.

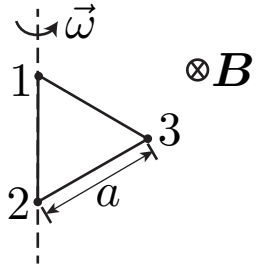


Figure 11.7: A rotating triangular loop.

3. A long solenoid with radius $r_1 = 5$ cm and $n = 1000$ turns per meter is oriented with its axis along the $\hat{z}$-direction. A current through the solenoid varies with time according to:

$$I(t) = I_0 \sin(\omega t),$$

where $I_0 = 2$ A, $\omega = 100\pi$ rad/s, and we define the counter-clockwise direction, as viewed from the $+z$ end of the axis, to be positive for the solenoid current, and likewise for the emf induced in the loop and for the current it carries.

A circular conducting loop of radius $r_2 = 3$ cm and resistance $R = 0.5\,\Omega$ is placed inside the solenoid, with its plane perpendicular to the solenoid's axis.

(a) Find the magnetic field $B(t)$ inside the solenoid as a function of time.

(b) Calculate the induced emf in the loop as a function of time.

(c) Calculate the maximum induced current in the loop.

(d) Sketch a graph showing $I(t)$ in the solenoid and the induced current in the loop over one complete period.

4. A uniform magnetic field is confined to a long cylindrical region of radius R. The field points out of the page and varies with time according to

$$B(t) = B_0 e^{-t/\tau},$$

where B_0 and τ are positive constants. There are no charges nearby, so the only electric field is the one induced by the changing $\mathbf{B}$.

(a) Find the magnitude of the induced electric field $E(r, t)$ at a distance r from the axis for both $r < R$ (inside the magnetic field region) and $r > R$ (outside the magnetic field region).

(b) Sketch the spatial dependence of $E(r)$ at a fixed time t. At what radius is the electric field maximized?

(c) A charged particle with charge q and mass m is released from rest at $t = 0$ at a distance r_0 from the axis, and is held in a frictionless insulating circular tube of radius r_0 so that its motion is confined to the $r = r_0$ orbit. Find the speed of the particle as $t \to \infty$, and the sense in which it then circulates as seen on the page (discuss both $r_0 < R$ and $r_0 > R$ cases).

5. A thin stationary conducting loop of resistance R lies in the xy-plane. It encloses the cross sections of two long solenoids parallel to the z axis, which are circles of radii a_1 and a_2 that do not overlap; the loop itself lies outside both solenoids. For $0 \le t \le T$ the fields inside the solenoids are

$$\mathbf{B}_1 = B_0 \frac{t}{T}\,\hat{z}, \qquad \mathbf{B}_2 = -B_0 \left(\frac{t}{T}\right)^2 \hat{z},$$

where $B_0 > 0$, $T > 0$ and $a_1^2 < 2a_2^2$, and outside the solenoids the magnetic field is zero. Neglect the self-inductance of the loop. Take $d\mathbf{S}$ along $+\hat{z}$, so that the emf and the current are positive when they circulate counterclockwise as seen from $+z$.

(a) Find the emf $\mathcal{E}(t)$ and the current $I(t)$ in the loop.

(b) Find the time t_* at which the current reverses, and give its sense before and after t_*. Explain both senses with Lenz's law.

(c) Find the net charge $Q = \int_0^T I\,dt$ that passes a given point of the loop, counted positive in the counterclockwise sense.

6. Two long, straight, parallel conducting rails lie in a horizontal plane a distance L apart and are joined at one end by a resistor R. A uniform magnetic field of magnitude B points vertically upward. A straight conducting rod of mass m lies across the rails, perpendicular to them, and slides along them without friction. At $t = 0$ the rod moves with speed v_0 directly away from the resistor; after that, no horizontal force acts on it except the magnetic one. Neglect the resistance of the rod and of the rails, and the self-inductance of the circuit.

(a) Find the current in the circuit when the rod's speed is v, and its sense as seen from above.

(b) Find the rod's speed $v(t)$ and the total distance it travels.

(c) Find the total energy dissipated in the resistor, and compare it with the rod's initial kinetic energy.

7. A thin conducting disk of radius a turns with constant angular speed ω about its axis, the z axis, counterclockwise as seen from $+z$, in a uniform magnetic field $\mathbf{B} = B\hat{z}$ with $B > 0$. One sliding contact touches the axle at the center of the disk and another touches its rim; the two are joined by a stationary external wire through a resistor R. Neglect all other resistance, all friction, and the self-inductance of the circuit.

(a) Using the magnetic force on a positive charge carried round with the disk, find the emf between the center and the rim, and say which of the two is at the higher potential.

(b) Find the current, the torque that the magnetic field exerts on the disk, and the torque that must be applied to keep ω constant. Show that the mechanical power supplied equals the power dissipated in R.

(c) The magnetic flux through the disk does not change as it turns. Consider instead, for $0 < t < 2\pi/\omega$, the closed path made of the external circuit, a radius drawn on the disk that passed the rim contact at $t = 0$ and turns with the disk, and the arc of the rim that this radius has swept since then, from the contact to the end of the radius. Find the rate at which the flux through this path changes, and compare it with your answer to (a).

8. A long solenoid of radius a, with n turns per unit length and its axis along z, carries a current $i(t) = kt$ with $k > 0$, flowing counterclockwise as seen from $+z$. Inside it $\mathbf{B} = \mu_0 nkt\,\hat{z}$, and outside it $\mathbf{B} = 0$. The induced electric field circles the axis, with magnitude $E = \mu_0 nkr/2$ for $r < a$ and $E = \mu_0 nka^2/2r$ for $r > a$, where r is the distance from the axis.

(a) Write $\mathbf{E}$ in Cartesian components, including its sense, and verify that $\nabla \times \mathbf{E} = -\partial\mathbf{B}/\partial t$ both inside and outside the solenoid.

(b) A circular wire loop of radius $b > a$ and resistance R lies in a plane perpendicular to the axis, centered on it. Neglecting its self-inductance, find the current in it and its sense as seen from $+z$, although $\mathbf{B} = 0$ at every point of the wire.

(c) Another loop lies outside the solenoid, in a plane perpendicular to the axis. It is made of two circular arcs of radii b_1 and b_2, with $a < b_1 < b_2$, centered on the axis and both subtending the angle α, with $0 < \alpha < 2\pi$, joined by two radial segments. Find its emf by integrating $\mathbf{E}$ around it, and compare the result with Faraday's law. Is $\mathbf{E}$ zero on this loop?

9. A long solenoid of radius b, with its axis along z, produces inside it the uniform field $\mathbf{B} = \beta t\,\hat{z}$ with $\beta > 0$, and no field outside. Work in the plane $z = 0$, far from the ends, with the origin on the axis. Assume that there are no charges, that the only electric field is the induced one, and that this field has the rotational symmetry of the solenoid and no component along z. Let $P = (a, 0)$ and $Q = (-a, 0)$, with $0 < a < b$.

(a) Find the induced electric field on the circle $r = a$, including its sense.

(b) Find $\int_P^Q \mathbf{E} \cdot d\mathbf{l}$ along the upper half of that circle ($y > 0$), along its lower half, and along the straight diameter, each traversed from P to Q.

(c) Can a potential V with $\mathbf{E} = -\nabla V$ be defined inside the solenoid? Relate the difference between the two semicircle integrals to Faraday's law.